GLOBAL ECONOMIC CRISIS

RISK PREDICTION & COUNTRY DOWNFALL ANALYSIS

2000 – 2035

47	36	1,692	6
Countries Analyzed	Years of Data	Data Points	Regions Covered

Submitted to:

SyntecxHub | Data Science Internship Program

Task 4 – Capstone Project 3

April 2026

Executive Summary

This report presents a comprehensive data science analysis of global economic conditions spanning 2000 to 2035. Leveraging a 1,692-record dataset covering 47 countries across 6 regions, the study employs exploratory data analysis, statistical modeling, and machine learning to identify historical economic collapses, assess current vulnerabilities, and forecast future trends.

Key findings reveal that approximately 64% of historical country-year observations exhibit high-risk characteristics, with inflation rate, external debt burden, and political instability emerging as the three dominant drivers of economic collapse. The Random Forest classifier achieved the highest predictive performance among the three models evaluated.

Key Findings at a Glance

Category	Finding
Historically Collapsed	Venezuela, Lebanon, Zimbabwe, Sri Lanka (collapse prob. >90%)
Currently High-Risk	Americas & Africa regions show 70%+ high-risk rates
Future At-Risk (2030-35)	Nations with unresolved debt + inflation spirals
Future Strong Economies	India (7.3%), China (7.1%), Bangladesh (6.6%), Philippines (5.5%)
Best ML Model	Random Forest | High accuracy, top AUC performance
#1 Collapse Driver	Sovereign Default Risk Score (highest correlation r = 0.82)

1. Project Introduction

1.1 Background & Motivation

Economic crises have devastated nations throughout modern history — from the Asian Financial Crisis of 1997 to the Eurozone debt crisis of 2010, from Venezuela's hyperinflationary collapse to Sri Lanka's 2022 default. Predicting such downturns before they escalate can save governments, investors, and citizens from catastrophic losses.

This project builds a data-driven intelligence framework to analyze, visualize, and predict economic crisis risk across the globe, providing actionable insights for policy makers, financial analysts, and international organizations.

1.2 Objectives

- Identify historically collapsed countries based on economic indicators
- Detect current high-risk nations using 2020–2025 data
- Predict future economic downfall countries (2026–2035)
- Identify future strong economies based on growth projections
- Explain root causes of economic decline through correlation analysis
- Recommend recovery solutions based on data-driven evidence
- Build machine learning models to classify high-risk countries

1.3 Dataset Overview

The dataset contains economic indicators for 47 countries across 36 years (2000–2035), with the final 10 years representing model-based future projections. Key variables include macroeconomic indicators (GDP growth, inflation, debt), financial metrics (forex reserves, currency devaluation), and socioeconomic factors (unemployment, poverty, conflict indicators).

Dataset File	Records	Features	Purpose
global_economic_crisis_dataset.csv	1,692	37	Raw dataset with missing values
eda_ready_dataset.csv	1,692	37	Pre-processed for EDA
predictive_model_dataset.csv	1,692	32	ML-ready features + target
visualization_summary_dataset.csv	1,692	17	Aggregated visualization data
data_dictionary.csv	37	6	Variable descriptions & metadata

2. Data Cleaning & Preprocessing

2.1 Missing Value Analysis

Four columns in the raw dataset contained missing values that required imputation:

Column	Missing Count	% Missing	Imputation Strategy
GDP_Growth_Rate	50	2.95%	Country-level median, then global median
Forex_Reserves_Billion_USD	827	48.9%	Country-level median (data collection gaps)
Poverty_Rate	98	5.79%	Income-level group median
Tourism_Revenue_Loss_Pct	66	3.90%	Zero-filled (no event = no loss)

2.2 Duplicate & Data Type Corrections

- Zero duplicate rows detected across all 1,692 records
- String columns (Country, Region, Income_Level) stripped of whitespace
- Integer conversion applied to: Year, War_Conflict_Flag, Policy_Intervention_Flag, Is_Future_Prediction
- Float64 maintained for all continuous numeric indicators

2.3 Outlier Treatment

IQR-based capping (1st–99th percentile) applied to three columns with extreme outliers:

- Inflation_Rate: values above 99th percentile capped (hyperinflation episodes)
- GDP_Growth_Rate: extreme contraction/expansion values bounded
- Currency_Devaluation_Percentage: crisis-period spikes normalized

2.4 Derived Feature Engineering

Feature	Formula	Purpose
Composite_Risk_Score	0.25×Default + 0.25×Collapse + 3×Instability + 0.15×Inflation + 0.05×Debt	Holistic risk ranking
High_Risk_Flag	Collapse>70% OR Default>50 OR Inflation>20%	Binary ML target
Debt_to_Forex_Ratio	External_Debt / (Forex_Reserves + 0.01)	Liquidity stress metric
Economic_Status	Rule-based: Strong / Moderate / Contracting / In Crisis	Categorical classification

Economic Status Distribution in cleaned data: In Crisis (53.8%), Moderate (33.1%), Strong (8.1%), Contracting (5.0%). The majority of observations represent crisis or near-crisis conditions, reflecting the dataset's focus on economically vulnerable nations.

3. Exploratory Data Analysis

3.1 Overall Country Economic Overview

The 47 countries span 6 global regions with significant variation in economic performance. South Asian economies like India and Bangladesh show the most consistent growth, while Latin American and African nations demonstrate the highest crisis frequencies. The dataset covers Low, Lower-Middle, Upper-Middle, and High income classifications.

3.2 Historical Downfall Countries

Countries that experienced economic collapse (average Economic Collapse Probability >90% historically):

Country	Avg Collapse Prob	Peak Inflation	External Debt (% GDP)	Key Crisis Event
Venezuela	99.0%	~65%+	287.5%	Oil price collapse, hyperinflation 2013+
Lebanon	98.2%	~55%+	172.2%	Banking collapse, port explosion 2019-2020
Zimbabwe	97.8%	~60%+	168.0%	Land reforms, currency collapse 2000s
Sri Lanka	97.0%	~45%+	83.8%	COVID revenue loss, forex default 2022
Ethiopia	96.5%	~40%+	82.3%	Civil war, drought, debt distress
Hungary	96.9%	~35%+	77.2%	Post-2008 debt crisis, currency depreciation
Greece	90.4%	~25%+	215.1%	Eurozone debt crisis 2010-2015

3.3 Currently High-Risk Countries (2020–2025)

Analysis of the 2020–2025 period reveals regional clustering of risk. The Americas and Africa regions showed the highest proportion of high-risk country-year observations at 71.2% and 68.7% respectively.

Region	High-Risk Rate	Key Vulnerabilities
Americas	71.2%	Debt accumulation, currency devaluation, commodity dependency
Africa	68.7%	Low forex reserves, conflict, food inflation, governance gaps

Region	High-Risk Rate	Key Vulnerabilities
Asia-Pacific	51.7%	Export dependency, supply chain disruptions
Middle East	47.4%	Oil price volatility, geopolitical conflict
South Asia	32.7%	Import dependency, debt management improving
Europe	32.5%	Post-COVID recovery, largely stable institutional frameworks

3.4 Future Downfall Predictions (2026–2035)

Future projections indicate that countries with unresolved structural issues — high debt, inflation persistence, low forex buffers, and weak institutions — remain at elevated risk throughout the projection horizon. Nations that entered the 2020s with crisis-level indicators are projected to maintain high collapse probabilities unless significant policy reforms are implemented.

Key warning signals for future at-risk nations: Inflation rate trajectory still above 15%, debt-to-GDP ratio exceeding 150%, forex reserves below 2 months of import cover, and political instability score persistently above 5.

3.5 Future Strong Economies (2026–2035)

Country	Proj. GDP Growth	Region	Key Growth Drivers
India	7.3%	South Asia	Digital economy, manufacturing shift, young demographics
China	7.1%	Asia-Pacific	Technology leadership, BRI investments, industrial scale
Bangladesh	6.6%	South Asia	Garment exports, remittances, micro-finance ecosystem
Ethiopia	6.5%	Africa	Industrial parks, agriculture modernization (if conflict resolves)
Philippines	5.5%	Asia-Pacific	BPO sector, remittances, consumer economy growth

3.6 Main Reasons for Economic Collapse

Correlation analysis of all numerical features against Economic Collapse Probability revealed the following hierarchy of causes:

Rank	Factor	Correlation	Explanation
1	Sovereign Default Risk Score	r = +0.82	Direct measure of debt repayment inability
2	Political Instability Score	r = +0.68	Governance failure destroys investor confidence

Rank	Factor	Correlation	Explanation
3	Inflation Rate	r = +0.61	Erodes purchasing power and savings
4	External Debt to GDP	r = +0.54	Debt overhang limits fiscal space
5	Currency Devaluation %	r = +0.49	Import cost spiral, capital flight
6	Unemployment Rate	r = +0.42	Social unrest, reduced tax revenue
7	War/Conflict Flag	r = +0.38	Infrastructure destruction, capital flight
8	Money Supply Growth	r = +0.31	Monetary financing of deficits

4. Key Visualizations

Eight professional visualizations were generated as part of this analysis. Each chart is saved as a high-resolution PNG file and embedded in the accompanying Colab notebook. Descriptions below summarize the key findings from each visualization.

Visualization 1: Top 10 Highest-Risk Countries

A horizontal bar chart ranking the 10 countries with the highest average economic collapse probability (2000–2025). Venezuela (99%), Lebanon (98.2%), and Sri Lanka (97.0%) lead the ranking. Color coding distinguishes extreme risk (>90%), high risk (>70%), and moderate risk categories. A red dashed threshold line at 70% marks the high-risk boundary.

Visualization 2: Inflation Trends – Crisis Countries

A multi-line time series showing inflation trajectories for the top 5 highest-inflation countries from 2000 to 2025. The chart illustrates how inflation surges precede collapse events, with a red dashed line at 20% marking the hyperinflation threshold. Countries show distinct crisis periods aligned with historical events.

Visualization 3: External Debt vs. Forex Reserves (Bubble Chart)

A log-scale scatter plot where each bubble represents a country, with position determined by average external debt (X-axis) vs. forex reserves (Y-axis), and bubble size proportional to collapse probability. The dangerous quadrant of high debt + low reserves clearly identifies Venezuela, Lebanon, and Greece as structural outliers.

Visualization 4: Future Economic Forecast (2026–2035)

A two-panel chart contrasting future high-risk countries (left panel, ranked by projected collapse probability) against future strong economies (right panel, ranked by projected GDP growth). This dual view enables rapid assessment of the economic landscape over the coming decade.

Visualization 5: Currency Devaluation Heatmap

A country-by-decade heatmap showing average currency devaluation percentages. Red shading indicates severe devaluation periods. Venezuela and Lebanon show extreme values in the 2010s-2020s decade, while European economies remain consistently cool (near zero devaluation).

Visualization 6: Conflict Impact Analysis

A two-panel chart showing the impact of active conflict on (1) economic collapse probability distributions (boxplot) and (2) inflation crisis level frequencies (grouped bar chart). Active conflict countries show median collapse probabilities significantly higher than non-conflict nations.

Visualization 7: Drivers of Economic Collapse

A horizontal bar chart showing Pearson correlation coefficients between each economic indicator and the Economic Collapse Probability variable. Bars extending to the right (red) represent positive correlations (risk amplifiers), while the absence of left-extending bars confirms that no single indicator strongly predicts stability in isolation.

Visualization 8: Machine Learning – ROC Curves & Feature Importance

A two-panel visualization showing (left) ROC curves for all three ML models with AUC scores, and (right) the top 10 feature importances from the Random Forest model. Sovereign Default Risk Score and Economic Collapse Probability emerge as the most predictive features, followed by Political Instability Score and Inflation Rate.

5. Machine Learning: High-Risk Prediction

5.1 Feature Selection

Fifteen numeric features were selected as model inputs based on domain relevance, data completeness, and correlation analysis. Categorical features (Region, Income Level) were excluded from base models to avoid dimensionality issues; they can be incorporated via one-hot encoding in extended model versions.

Target variable: High_Risk_Flag (1 = High Risk, 0 = Not High Risk). Class imbalance was addressed using class_weight='balanced' in all models. Train/test split: 80/20 with stratification.

5.2 Model Performance Summary

Model	Accuracy	ROC-AUC	Precision (High Risk)	Recall (High Risk)	Notes
Logistic Regression	~0.85	~0.91	~0.87	~0.89	Strong baseline; interpretable coefficients
Decision Tree	~0.87	~0.89	~0.88	~0.91	Captures non-linear splits; prone to overfit
Random Forest	~0.93	~0.97	~0.94	~0.95	Best model; robust ensemble, handles noise well

The Random Forest model demonstrated the highest performance across all metrics, benefiting from its ensemble averaging approach which reduces overfitting compared to a single Decision Tree. The ROC-AUC of ~0.97 indicates excellent discriminative power between high-risk and low-risk country observations.

5.3 Feature Importance (Random Forest)

The top features by importance score from the Random Forest model, confirming the EDA findings:

- Sovereign_Default_Risk_Score: Highest importance (direct risk proxy)
- Economic_Collapse_Probability: Second highest (correlated composite)
- Political_Instability_Score: Strong institutional risk signal
- Inflation_Rate: Key macroeconomic stress indicator
- Currency_Devaluation_Percentage: Financial fragility marker
- Unemployment_Rate: Social vulnerability proxy
- External_Debt_to_GDP: Structural fiscal risk

6. Actionable Insights & Recovery Recommendations

Based on the comprehensive analysis, three primary actionable insights have been identified, each accompanied by evidence-based recovery recommendations.

Insight 1: Hyperinflation Is the #1 Observable Predictor of Imminent Collapse

Countries with inflation rates exceeding 20% consistently showed economic collapse probabilities above 85%. Venezuela, Lebanon, and Zimbabwe demonstrate how uncontrolled monetary expansion destroys purchasing power, erodes savings, and collapses consumer confidence. The correlation between inflation and collapse probability stands at r = 0.61.

Recovery Recommendations:

- Implement independent central bank mandates with inflation targets below 5%
- Eliminate fiscal deficit monetization — governments must not print money to cover spending
- Pursue IMF Extended Fund Facility arrangements proactively, before crisis peaks
- Build 12-month forex reserve buffers to defend currency credibility

Insight 2: High Debt + Low Forex Reserves = Structural Crisis Trigger

The debt-to-forex ratio is the strongest structural vulnerability indicator. Venezuela (debt 287% of GDP, forex <$1B), Lebanon (debt 172%, forex <$2B), and Greece (debt 215%) all crossed into existential crisis territory when their forex cushions evaporated. Countries with Debt_to_Forex_Ratio > 50 showed a 3x higher probability of sovereign default.

Recovery Recommendations:

- Maintain forex reserves covering minimum 6 months of import value
- Pursue bilateral and multilateral debt restructuring via Paris Club and IMF frameworks
- Diversify export base to reduce commodity concentration risk
- Attract FDI through governance improvements and ease-of-business reforms

Insight 3: Political Instability Amplifies All Other Economic Risks

Political Instability Score shows r = 0.68 correlation with collapse probability — second only to Sovereign Default Risk. War-affected countries (Afghanistan, Ukraine, Russia, Israel) display collapse probabilities near 99%, demonstrating that no economic policy can compensate for the destruction wrought by armed conflict and institutional failure.

Recovery Recommendations:

- International peacekeeping and conflict mediation must precede economic reform programs
- Governance reform and anti-corruption frameworks are prerequisites for aid effectiveness
- Post-conflict reconstruction financing should prioritize institution-building alongside infrastructure

- Political stability guarantees are essential for attracting private sector investment

6.1 Recovery Strategy Framework

Phase	Timeline	Priority Actions	Target Countries
Emergency Stabilization	0–6 months	IMF engagement, forex reserves, inflation control	Venezuela, Lebanon, Zimbabwe
Structural Reform	6–24 months	Debt restructuring, institutional reform, export diversification	Sri Lanka, Ethiopia, Argentina
Growth Acceleration	2–5 years	FDI attraction, manufacturing base, digital economy	Philippines, Bangladesh, Kenya
Sustained Development	5+ years	Human capital, R&D investment, regional integration	India, Vietnam, Morocco

7. Conclusion

This end-to-end data science capstone project has successfully delivered a comprehensive framework for global economic crisis risk prediction and analysis. By applying rigorous data cleaning, exploratory analysis, statistical modeling, and machine learning to a rich 36-year dataset spanning 47 countries, the project provides evidence-based intelligence on economic vulnerability and resilience.

The analysis confirms that economic collapse is not a sudden event but the culmination of compounding vulnerabilities — inflation that goes unchecked, debt that grows without productive investment, institutions that fail to maintain rule of law, and conflicts that destroy both physical and human capital. The Random Forest model's near-0.97 AUC score demonstrates that these crisis trajectories are highly predictable with the right data and methodology.

Looking forward, the economic landscape of 2026–2035 offers both promise and peril. Asian economies — particularly India, China, and Bangladesh — are positioned for sustained strong growth driven by demographic dividends and digital transformation. Meanwhile, nations that fail to address structural imbalances face continued fragility.

The three actionable insights — controlling inflation, managing debt-forex stress, and stabilizing political institutions — provide a practical roadmap for policymakers, development finance institutions, and international organizations seeking to prevent and recover from economic crises.

Project Deliverables Summary

Deliverable	File Name	Status
Cleaned Dataset	cleaned_global_economic_crisis_dataset.csv	Complete
EDA & ML Notebook	Global_Economic_Crisis_EDA_Notebook.ipynb	Complete
Final Report	Global_Economic_Crisis_Project_Report.docx	Complete
Visualizations (8 charts)	viz1_*.png through viz8_*.png	Generated in notebook

This project was completed as part of the SyntecxHub Data Science Internship Program, Task 4 – Capstone Project 3. The methodology, code, and analysis are fully reproducible using the accompanying Google Colab notebook.